

LPDDR5X PAVES THE WAY

Didn't we just get LPDDR5 on smartphones? Here's why the newer LPDDR5X standard matters

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LPDDR5X is the latest memory standard for low power applications such as smartphones and Micron's DRAM was just validated for MediaTek's new Dimensity 9000 chipset. We spoke with Raj Talluri, Senior VP and GM of the Mobile Business Unit at Micron to understand what the fuss is all about the new memory standard.

Q What was the impact of the move from the '1z' Node to the '1α' (1-alpha) Node? On the consumer desktop side, the move from DDR4 to DDR5 which shows significant performance gains in synthetic benchmarks shows not much real world performance gains in practical application.

Raj: Basically when you go into a new node, a couple of good things happen. One, because of the shrink we are able to get more die per wafer which helps with the cost. We are able to be more cost competitive because as the memory density requirement keeps going up because of the use cases, if we don't shrink the nodes the phones will become cost prohibitive. So we work on reducing the cost by shrinking. The other one is whenever you shrink, the distance the signal has to travel becomes lesser. So you are able to get higher performance and you are able to get lower power. Those are usually



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the advantages i.e. lower power, higher performance and better cost structure. We see all of that in our 1α Node.

Now about your question about when technology transitions from LPDDR4 TO LPDDR5, LPDDR5 TO LPDDR5X and so on. As I mentioned previously, I have been in this business for long time and see the industry move from LPDDR2 to LPDDR3 to LPDDR4 to LPDDR5 or from 2G to 3G to 4G and now to 5G. What typically happens is that as a component supplier, we pretty much invest first of all the people that invest in the ecosystem. We make high performance or high bandwidth memory possible. Then the processor companies need to build the processors to take advantage of that so they're able to run at that high speed and connect to it. And then the operating system developers have to take advantage of the higher bandwidth. And then the applications have to come.

Let's say, for example, that 8K resolution video starts streaming to

your phone. Right now you can only get 4K, you cannot stream 8K. And many times people ask why you need 8K? Because you know, the same question was asked at the advent of 4K. Typically what happens is, as the performance of the network increases and as the performance of the processor increases you'll see the applications come. And when the application comes, you will see the advantage. Typically the benchmarks are kind of like a harbinger of what is to come. What's in the benchmarks today will be in the applications of tomorrow. And the best applications take time to develop because for people to write applications that take advantage of it, there has to be enough of an install base of phones. So there has to be millions of phones for someone to write an app that takes advantage of a 5G use case. And the infrastructure has to build up so typically there is a lag in the industry from the time we make the product, to when our customers make their products, to the time of launch in the market, to the operating system catching up and then the applications catch up. And when that happens you'll wonder, "How were we able to live with the previous generation?"

Q Very true. Since LPDDR5X is primarily marketed towards phones, that's where I wanted to understand what kind of applications can we foresee that can make the best use of the improved bandwidth and latency of LPDDR5X?

Raj: This is the virtuous cycle. I'll make up an example. So let's say you are at a sports stadium and watching a sports video. On the user side, typically there is a technology which comes into play when there is an action of something happening, such as a player getting bowled out or a player throwing a ball. Immediately, you get a 360-degree